

Brain tumors are often treated with radiation consisting of high energy photons, like gamma rays, that kill cancer cells through apoptosis, autophagy, DNA damage, etc. Local drug delivery provides a high concentration of drug in the tumor while decreasing systemic toxicity.

Researchers study a new drug that emits gamma rays at 50×10^{18} Hz and is injected into tumors using a vascular catheter and x-ray imaging guidance. The x-rays have a lower frequency, 100×10^{15} Hz, compared to gamma rays. The catheter also collects tumor samples for optical analysis using lenses, as shown in Figure 1.

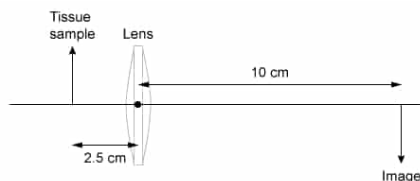


Figure 1 A tissue sample placed 2.5 cm from the center of a convex lens forms an image 10 cm from the lens

An x-ray tube propels high energy electrons at a tungsten target to produce x-rays but also heats the target. To avoid overheating, tube current I is applied only to collect one x-ray image, followed by a cooling period (Figure 2). The tungsten target is 1000°C and 2 cm long after cooling, but it heats to 2000°C and increases in length by 0.1 mm when generating x-rays.

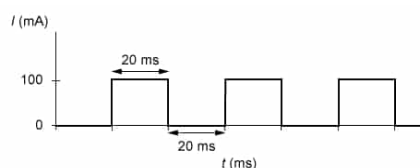


Figure 2 X-ray tube current over time

The drug is injected into a blood vessel feeding the tumor with the injection syringe and catheter at the same level to eliminate gravitational effects. In one experiment, two drug doses were compared and the higher drug dose had a greater therapeutic response. In another experiment, the drug injection velocity was varied and the lowest injection velocity had the greatest therapeutic response.

If the density of the drug is known, what additional information is required to calculate the velocity of the drug as it enters the blood vessel? (Note: Assume ideal fluid flow.)

- ☐ A. Pressure in syringe, pressure in blood vessel, and mass of drug [16%]
- ☒ B. Pressure in syringe, velocity in syringe, and pressure in blood vessel [57%]
- ☐ C. Pressure in syringe, velocity in syringe, and volume of drug [19%]
- ☐ D. Velocity in syringe, mass of drug, and volume of drug [6%]

Correct

56% answered correctly

Explanation:

Venturi effect during drug injection into vessel

$$P_A + \frac{1}{2} \rho v_A^2 = P_B + \frac{1}{2} \rho v_B^2$$

$$v_B = \sqrt{\frac{2(P_A - P_B)}{\rho} + v_A^2}$$

- P_A Pressure in syringe
- P Density
- v_A Velocity in syringe
- P_B Pressure in vessel
- v_B Velocity in vessel

Calculating v_B requires pressure in syringe, velocity in syringe, and pressure in vessel.

Bernoulli's equation describes the flow of an **ideal fluid** within a pipe from point A to point B. At each of these two points, the total energy of the fluid remains constant. Bernoulli's equation includes three energy terms, **pressure P** , **potential energy**, and **kinetic energy**.

Bernoulli's principle is derived from Bernoulli's equation under the special condition of constant height for the fluid.

Hence, the potential energy terms cancel, and the equation depends only on P , fluid density ρ , and velocity v :

$$P_A + \frac{1}{2}\rho v_A^2 = P_B + \frac{1}{2}\rho v_B^2$$

This equation describes the [Venturi effect](#), in which decreased P is associated with increased v (and vice versa) to maintain a constant sum of the energy terms. Furthermore, solving for the velocity at point B v_B yields:

$$v_B = \sqrt{\frac{2(P_A - P_B)}{\rho} + v_A^2}$$

In this question, the ρ of the drug is known and the researchers want to calculate v_B . Therefore, the pressure in the syringe P_A , velocity in the syringe v_A , and pressure in the blood vessel P_B are required to calculate v_B .

(Choices A and D) The mass of the drug is not needed to calculate velocity because it is not included in any of the energy terms for fluid flow.

(Choices C and D) The volume of the drug is not needed to calculate velocity because it is not included in any of the energy terms for fluid flow.

Educational objective:

The Venturi effect describes the decreased pressure associated with increased velocity of a fluid flowing in a pipe. This effect derives from a special case of Bernoulli's equation, in which the height of the fluid remains constant.

Time spent:0 sec

Subject:
Physics

QID:403751

System:
Fluids

Brain tumors are often treated with radiation consisting of high energy photons, like gamma rays, that kill cancer cells through apoptosis, autophagy, DNA damage, etc. Local drug delivery provides a high concentration of drug in the tumor while decreasing systemic toxicity.

Researchers study a new drug that emits gamma rays at 50×10^{18} Hz and is injected into tumors using a vascular catheter and x-ray imaging guidance. The x-rays have a lower frequency, 100×10^{15} Hz, compared to gamma rays. The catheter also collects tumor samples for optical analysis using lenses, as shown in Figure 1.

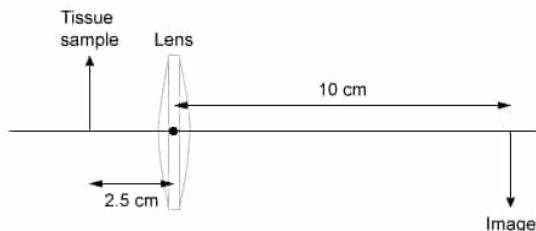


Figure 1 A tissue sample placed 2.5 cm from the center of a convex lens forms an image 10 cm from the lens

An x-ray tube propels high energy electrons at a tungsten target to produce x-rays but also heats the target. To avoid overheating, tube current I is applied only to collect one x-ray image, followed by a cooling period (Figure 2). The tungsten target is 1000°C and 2 cm long after cooling, but it heats to 2000°C and increases in length by 0.1 mm when generating x-rays.

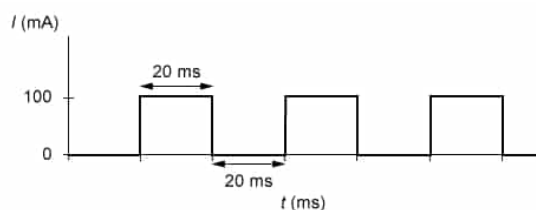


Figure 2 X-ray tube current over time

The drug is injected into a blood vessel feeding the tumor with the injection syringe and catheter at the same level to eliminate gravitational effects. In one experiment, two drug doses were compared and the higher drug dose had a greater therapeutic response. In another experiment, the drug injection velocity was varied and the lowest injection velocity had the greatest therapeutic response.

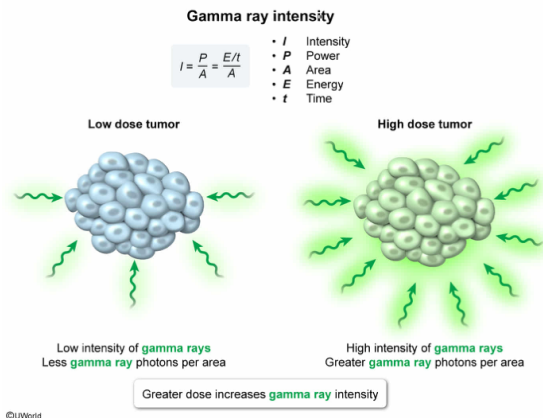
Which of the following statements best explains why the higher dose of gamma-emitting drug has a greater therapeutic response?

- ☒ A. The intensity of the gamma rays is greater. [40%]
- ☐ B. The frequency of the gamma rays is greater. [25%]
- ☐ C. The velocity of the gamma rays is faster. [2%]
- ☐ D. The energy of the gamma rays is greater. [30%]

Correct

40% answered correctly

Explanation:



The **intensity** I of **electromagnetic radiation**, such as light, is defined as the amount of power P delivered per unit area A :

$$I = \frac{P}{A}$$

Since P equals the ratio of energy E and time t , I can also be expressed as:

$$I = \frac{\frac{E}{t}}{A}$$

Therefore, the intensity of electromagnetic radiation is proportional to the individual energy of each emitted photon and the **number of photons** n emitted per unit time:

$$I \propto n$$

In this question, the drug emits gamma ray photons with a fixed frequency, velocity, and energy. Increasing the dose increases the number of photons emitted, but the frequency, velocity, and energy of each photon remain constant.

(Choices B, C, and D) The frequency, velocity, and energy of the gamma rays remain constant. Only the gamma ray intensity increases when the dose increases.

Educational objective:

The intensity of electromagnetic radiation is the amount of power, or energy per unit time, delivered per unit area. The gamma ray intensity increases with increased drug dose because more gamma ray photons are emitted.

Time spent: 0 sec

Subject:
Physics

QID: 403752

System:
Fluids

Brain tumors are often treated with radiation consisting of high energy photons, like gamma rays, that kill cancer cells through apoptosis, autophagy, DNA damage, etc. Local drug delivery provides a high concentration of drug in the tumor while decreasing systemic toxicity.

Researchers study a new drug that emits gamma rays at 50×10^{18} Hz and is injected into tumors using a vascular catheter and x-ray imaging guidance. The x-rays have a lower frequency, 100×10^{15} Hz, compared to gamma rays. The catheter also collects tumor samples for optical analysis using lenses, as shown in Figure 1.

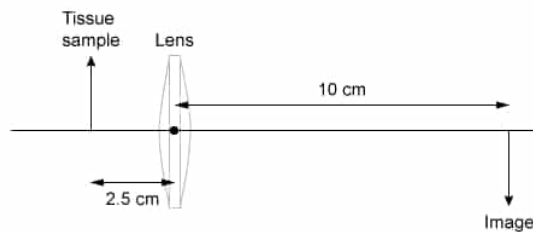


Figure 1 A tissue sample placed 2.5 cm from the center of a convex lens forms an image 10 cm from the lens

An x-ray tube propels high energy electrons at a tungsten target to produce x-rays but also heats the target. To avoid overheating, tube current I is applied only to collect one x-ray image, followed by a cooling period (Figure 2). The tungsten target is 1000°C and 2 cm long after cooling, but it heats to 2000°C and increases in length by 0.1 mm when generating x-rays.

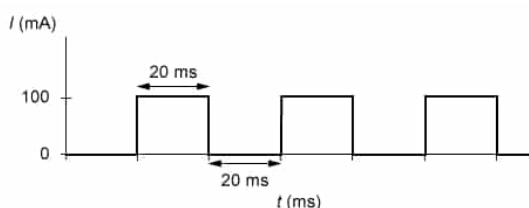


Figure 2 X-ray tube current over time

The drug is injected into a blood vessel feeding the tumor with the injection syringe and catheter at the same level to eliminate gravitational effects. In one experiment, two drug doses were compared and the higher drug dose had a greater therapeutic response. In another experiment, the drug injection velocity was varied and the lowest injection velocity had the greatest therapeutic response.

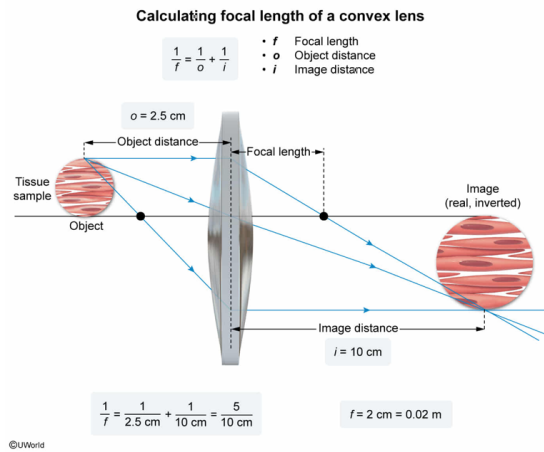
What is the focal length of the lens shown in Figure 1 of the passage?

- ☐ A. 0.1 m [13%]
- ☐ B. 0.08 m [7%]
- ☐ C. 0.04 m [17%]
- ☒ D. 0.02 m [61%]

Correct

61% answered correctly

Explanation:



In a **convex lens**, also called a converging lens, both surfaces curve outward. The **focal length** f of a converging lens is the distance from the lens to the point where parallel incident light rays converge. The **thin lens equation** states that f is related to the object distance o and image distance i by:

$$\frac{1}{f} = \frac{1}{o} + \frac{1}{i}$$

In this question, o equals 2.5 cm and i equals 10 cm. Substituting these values into the equation above yield:

$$\frac{1}{f} = \frac{1}{2.5 \text{ cm}} + \frac{1}{10 \text{ cm}} = \frac{4}{10 \text{ cm}} + \frac{1}{10 \text{ cm}} = \frac{5}{10 \text{ cm}}$$

Hence, the value of f is given by:

$$f = \frac{10 \text{ cm}}{5} = 2 \text{ cm}$$

Therefore, f equals 2 cm or 0.02 m.

(Choice A) 0.1 m results from assuming that the focal length equals the distance to the image.

(Choice B) 0.08 m is obtained by calculating the inverse of the sum of the object distance and image distance without converting the value to meters.

(Choice C) 0.04 m is calculated from the ratio of the image distance to the object distance.

Educational objective:

The focal length of a thin lens is calculated from the object distance and image distance using the thin lens equation: $1/f = 1/o + 1/i$.

Time spent: 0 sec

Subject:
Physics

QID: 403753

System:
Fluids

Brain tumors are often treated with radiation consisting of high energy photons, like gamma rays, that kill cancer cells through apoptosis, autophagy, DNA damage, etc. Local drug delivery provides a high concentration of drug in the tumor while decreasing systemic toxicity.

Researchers study a new drug that emits gamma rays at 50×10^{18} Hz and is injected into tumors using a vascular catheter and x-ray imaging guidance. The x-rays have a lower frequency, 100×10^{15} Hz, compared to gamma rays. The catheter also collects tumor samples for optical analysis using lenses, as shown in Figure 1.

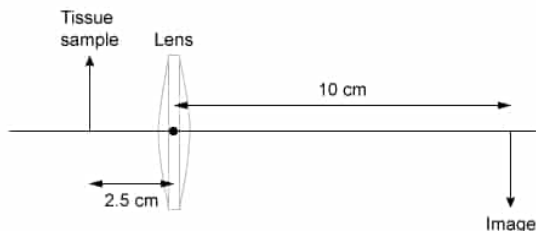


Figure 1 A tissue sample placed 2.5 cm from the center of a convex lens forms an image 10 cm from the lens

An x-ray tube propels high energy electrons at a tungsten target to produce x-rays but also heats the target. To avoid overheating, tube current I is applied only to collect one x-ray image, followed by a cooling period (Figure 2). The tungsten target is 1000°C and 2 cm long after cooling, but it heats to 2000°C and increases in length by 0.1 mm when generating x-rays.

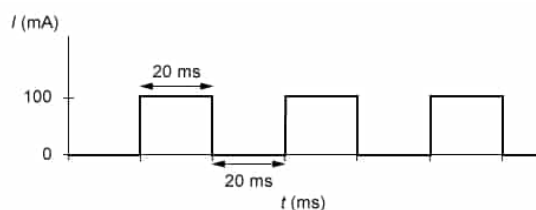
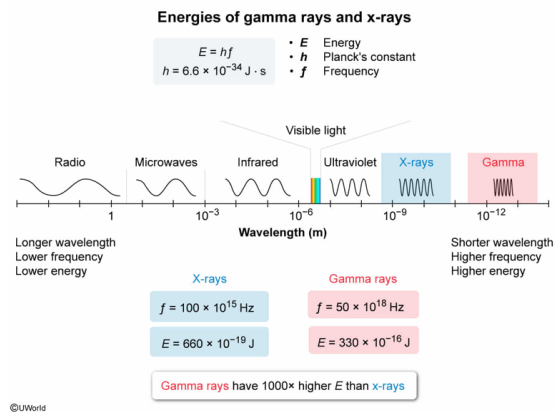


Figure 2 X-ray tube current over time

The drug is injected into a blood vessel feeding the tumor with the injection syringe and catheter at the same level to eliminate gravitational effects. In one experiment, two drug doses were compared and the higher drug dose had a greater therapeutic response. In another experiment, the drug injection velocity was varied and the lowest injection velocity had the greatest therapeutic response.

The advantage of using gamma rays for tumor therapy over x-rays is that gamma rays:

- ☐ A. are not electromagnetic radiation. [3%]
- ☐ B. are photons. [7%]
- ☒ C. have higher energy. [81%]
- ☐ D. have a longer wavelength. [6%]

Explanation:

Electromagnetic radiation includes different types of wave phenomena, such as radio waves, visible light, x-rays, and gamma rays. These different categories of waves are often organized into the **electromagnetic spectrum**, ordered by decreasing **wavelength** λ and increasing **frequency** f . The energy E of electromagnetic radiation equals the product of Planck's constant h and f .

$$E = hf$$

Hence, E is **directly proportional** to f .

In this question, f of the gamma rays described in the passage is $50 \times 10^{18} \text{ Hz}$, and f of the x-rays is $100 \times 10^{15} \text{ Hz}$. Because f is higher for gamma rays than x-rays, E will also be higher for gamma rays than for x-rays.

Therefore, the gamma rays used for tumor therapy have a higher E than the x-rays used for imaging.

(Choices A and B) Both x-rays and gamma rays are electromagnetic radiation composed of photons.

(Choice D) The wavelength of gamma rays is shorter than the wavelength of x-rays, not longer.

Educational objective:

The electromagnetic spectrum consists of different types of radiation ordered by wavelength, frequency, or energy. Electromagnetic radiation with a shorter wavelength has higher frequency and energy.

Time spent: 0 sec

Subject:
Physics

QID: 403754

System:
Fluids

Brain tumors are often treated with radiation consisting of high energy photons, like gamma rays, that kill cancer cells through apoptosis, autophagy, DNA damage, etc. Local drug delivery provides a high concentration of drug in the tumor while decreasing systemic toxicity.

Researchers study a new drug that emits gamma rays at 50×10^{18} Hz and is injected into tumors using a vascular catheter and x-ray imaging guidance. The x-rays have a lower frequency, 100×10^{15} Hz, compared to gamma rays. The catheter also collects tumor samples for optical analysis using lenses, as shown in Figure 1.

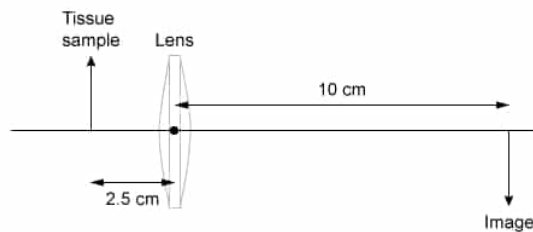


Figure 1 A tissue sample placed 2.5 cm from the center of a convex lens forms an image 10 cm from the lens

An x-ray tube propels high energy electrons at a tungsten target to produce x-rays but also heats the target. To avoid overheating, tube current I is applied only to collect one x-ray image, followed by a cooling period (Figure 2). The tungsten target is 1000°C and 2 cm long after cooling, but it heats to 2000°C and increases in length by 0.1 mm when generating x-rays.

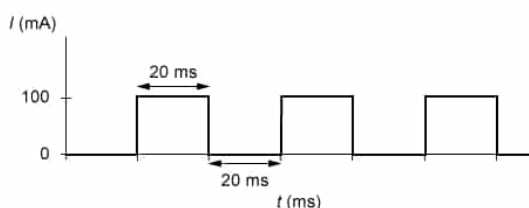


Figure 2 X-ray tube current over time

The drug is injected into a blood vessel feeding the tumor with the injection syringe and catheter at the same level to eliminate gravitational effects. In one experiment, two drug doses were compared and the higher drug dose had a greater therapeutic response. In another experiment, the drug injection velocity was varied and the lowest injection velocity had the greatest therapeutic response.

What is the coefficient of expansion for the tungsten target described in the passage?

- ☐ A. $2 \times 10^{-6} \frac{1}{^\circ\text{C}}$ [26%]
- ☒ B. $5 \times 10^{-6} \frac{1}{^\circ\text{C}}$ [40%]
- ☐ C. $20 \times 10^{-6} \frac{1}{^\circ\text{C}}$ [23%]
- ☐ D. $50 \times 10^{-6} \frac{1}{^\circ\text{C}}$ [8%]

Correct

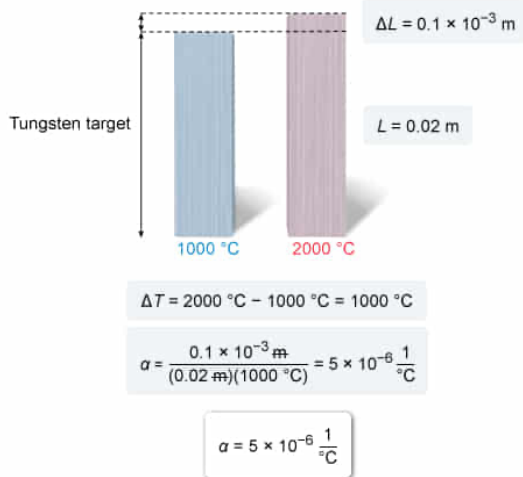
40% answered correctly

Explanation:

Coefficient of expansion for tungsten target

$$\Delta L = \alpha L \Delta T$$
$$\alpha = \frac{\Delta L}{L \Delta T}$$

- ΔL Change in length
- α Coefficient of expansion
- L Length
- ΔT Change in temperature



©UWorld

Metal objects expand when they are heated. The **change in length** ΔL of the object equals the product of the **coefficient of expansion** α for the material, the original **length** L of the object, and the **change in temperature** ΔT :

$$\Delta L = \alpha L \Delta T$$

Hence, ΔL is **directly proportional** to ΔT with a proportionality constant equal to α .

Solving this equation for α yields:

$$\alpha = \frac{\Delta L}{L \Delta T}$$

In this question, the tungsten target L is 2 cm and its temperature is 1000 °C during the cooling period. When generating x-rays, the target temperature increases to 2000 °C and ΔL is 0.1 mm. The difference between the temperatures while generating x-rays and during the cooling period equals ΔT :

$$\Delta T = 2000^\circ\text{C} - 1000^\circ\text{C} = 1000^\circ\text{C}$$

Substituting these values into the equation for α gives:

$$\alpha = \frac{(0.1 \text{ mm})}{(2 \text{ cm})(1000^\circ\text{C})}$$

Converting the length units to meters and ΔT to scientific notation yields:

$$\alpha = \frac{(0.1 \times 10^{-3} \text{ m})}{(2 \times 10^{-2} \text{ m})(10^3^\circ\text{C})} = 0.05 \times 10^{-3+2-3} \frac{1}{^\circ\text{C}} = 5 \times 10^{-6} \frac{1}{^\circ\text{C}}$$

Therefore, α for the tungsten target is $5 \times 10^{-6} \text{ 1/}^\circ\text{C}$.

(Choice A) $2 \times 10^{-6} \text{ } 1/^{\circ}\text{C}$ is obtained by using 0.2 mm for the change in length and 0.1 m for the length.

(Choice C) $20 \times 10^{-6} \text{ } 1/^{\circ}\text{C}$ results from calculating the ratio of the length and the change in temperature.

(Choice D) $50 \times 10^{-6} \text{ } 1/^{\circ}\text{C}$ is calculated by using 1 mm for the change in length instead of 0.1 mm.

Educational objective:

The linear relationship between an object's length and temperature depends on the coefficient of expansion. The coefficient of expansion can be calculated from the lengths of an object at two different temperatures.

Time spent:0 sec

Subject:
Physics

QID:403755

System:
Fluids

Brain tumors are often treated with radiation consisting of high energy photons, like gamma rays, that kill cancer cells through apoptosis, autophagy, DNA damage, etc. Local drug delivery provides a high concentration of drug in the tumor while decreasing systemic toxicity.

Researchers study a new drug that emits gamma rays at 50×10^{18} Hz and is injected into tumors using a vascular catheter and x-ray imaging guidance. The x-rays have a lower frequency, 100×10^{15} Hz, compared to gamma rays. The catheter also collects tumor samples for optical analysis using lenses, as shown in Figure 1.

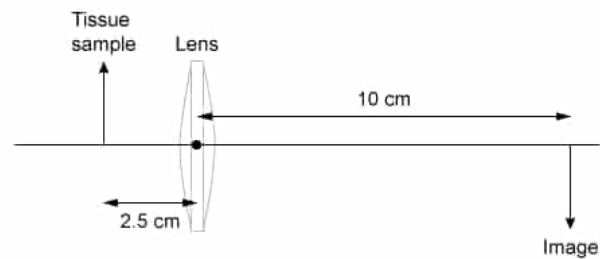


Figure 1 A tissue sample placed 2.5 cm from the center of a convex lens forms an image 10 cm from the lens

An x-ray tube propels high energy electrons at a tungsten target to produce x-rays but also heats the target. To avoid overheating, tube current I is applied only to collect one x-ray image, followed by a cooling period (Figure 2). The tungsten target is 1000 °C and 2 cm long after cooling, but it heats to 2000 °C and increases in length by 0.1 mm when generating x-rays.

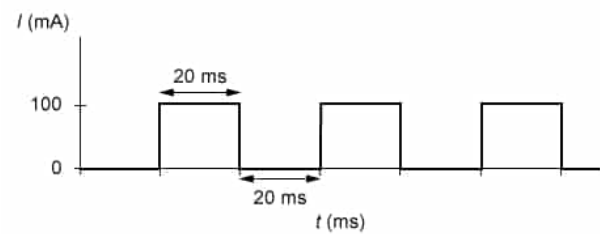


Figure 2 X-ray tube current over time

The drug is injected into a blood vessel feeding the tumor with the injection syringe and catheter at the same level to eliminate gravitational effects. In one experiment, two drug doses were compared and the higher drug dose had a greater therapeutic response. In another experiment, the drug injection velocity was varied and the lowest injection velocity had the greatest therapeutic response.

According to Figure 2 in the passage, how much charge flows into the x-ray tube during a single image?

- ☐ A. 100 mC [13%]
- ☐ B. 20 mC [26%]
- ☐ C. 10 mC [13%]
- ✓ ☒ D. 2 mC [46%]

Correct

46% answered correctly

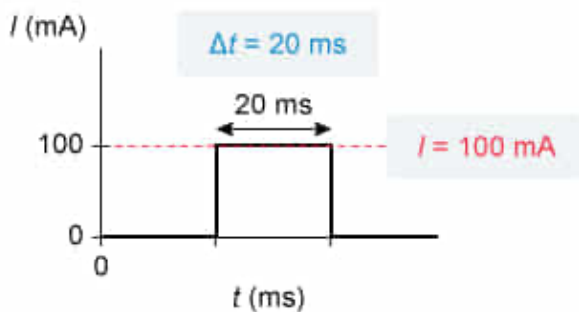
Explanation:

Charge flowing into x-ray tube

$$I = \frac{Q}{\Delta t}$$

$$Q = I \cdot \Delta t$$

- I Current
- Q Charge
- Δt Time interval



$$Q = (0.1 \text{ A})(0.02 \text{ s}) = 0.002 \text{ C} = 2 \text{ mC}$$

©UWorld

Electric current I equals the electric charge Q flowing through an electric circuit over a specific **time duration** Δt . According to **circuit sign conventions**, I is positive in the direction of the flow of positive Q and equals the ratio of Q and Δt :

$$I = \frac{Q}{\Delta t}$$

The unit for I is the ampere (A), which equals a Q of 1 Coulomb (C) flowing through the circuit in 1 s.

Furthermore, this equation can be rearranged to solve for Q :

$$Q = I \cdot \Delta t$$

In this question, I flowing into the x-ray tube equals 100 mA, which is 0.1 A, and Δt for a single x-ray image is 20 ms, which is 0.02 s. Substituting these values into the above equation yields:

$$Q = (0.1 \text{ A})(0.02 \text{ s}) = \left(0.1 \frac{\text{C}}{\text{s}}\right)(0.02 \text{ s})$$

$$Q = 0.002 \text{ C} = 2 \text{ mC}$$

Therefore, 2 mC flows into the x-ray tube for each x-ray image.

(Choice A) 100 mC is obtained by assuming that the charge equals the current.

(Choice B) 20 mC is calculated by using 0.2 s for the time interval.

(Choice C) 10 mC results from using 0.1 s for the time interval.

Educational objective:

The electric current equals the rate of charge flowing through an electric circuit.

Electric charge is calculated from the ratio of current and the interval of time.

Time spent:0 sec

Subject:

Physics

QID:403756

System:

Fluids